

Balanced Carrier Gate (BCG): admit only orientation-robust response channels

computed before ε_i and w_i are consulted

A

Two-sided response

usable for either keyed orientation
RRB sign-mass partition of $p_i(\text{base})$



$g_i(z) < 0$

$g_i(z) > 0$

$$S_i = 2 \min(.53, .47) = .94 > s_{\min} = .50$$

BCG admits

key asks +
one-draw mass = 0.53

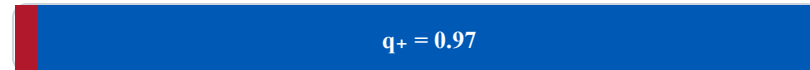
key asks -
one-draw mass = 0.47

RAR may steer • RRV counts one Bernoulli trial

B

One-sided response

usable only for one orientation
RRB sign-mass partition of $p_i(\text{base})$



$g_i(z) < 0$

$g_i(z) > 0$

$$S_i = 2 \min(.97, .03) = .06 < s_{\min} = .50$$

BCG rejects

key asks +
one-draw mass = 0.97

$P_8 = 1.00$

key asks -
one-draw mass = 0.03

$P_8 = 0.22$

reference sampling • no detector trial

Orientation-free gate: S_i is invariant to swapping the two RRB arms; ε_i and w_i enter only after admission.